

SPATIAL TAXONOMY · FLOOR PLATE METHODOLOGY

Key Plans and Tiles

A geometric self-similar aperiodic space-planning system. Furniture-driven Key Plans aggregate into Tiles; Tiles compose Floor Plates without remainder.
Methodology v12.

AGENDA — SEVEN SECTIONS

- 01 What Key Plans and Tiles Means
- 02 The Tile Catalogue — Small Tiles
- 03 The Tile Catalogue — Medium & Large Tiles
- 04 The Size Hierarchy & Floor Plate Range
- 05 One Floor — Filled Exactly
- 06 Design Principles
- 07 Leasing Economics & Rolling Efficiency

What “Key Plans and Tiles” Means

WORKING DEFINITION, FROM THE METHODOLOGY MEMO

“**Key Plans and Tiles**” means a geometric self-similar aperiodic space-planning system based on furniture/equipment arrangements and circulation versus modular area-per-person progressions. Various Key Plans are summed up together in order to make Tiles. Not all the Key Plans are supposed to fit in all the Tiles — in the case of the building corners or areas surrounding the Elevator Lobbies, there may be “unique” Special Tiles where either only certain Key Plans fit or the Key Plans found within these Special Tiles are exclusive to these areas. However, in all cases, the entirety of the Floor Plates is dividable by a certain combination of Key Plans, with no remainder of “free space” that is not represented by a Key Plan.

Translate: a Key Plan is the smallest unit — a labelled rectangle around one tenant's actual furniture, sized to enclose desks, storage, circulation, and accessibility. A Tile is a composable group of Key Plans. The Floor Plate is the assembly of Tiles. Nothing on the floor is unrepresented.

The system is *aperiodic* because Tiles do not repeat on a fixed grid. It is *self-similar* because each Tile is exactly one HVAC climate zone, regardless of size. It is *geometric* because all dimensions resolve to real metric measurements driven by real furniture SKUs.

The Tile Catalogue — Small Tiles

Small Tile Family — 2,700 SF (250.84 m²) each. Each Tile is one HVAC climate zone; combinations aggregate climate-control authority. Tile B-1 and Mixed E-1/E-2 carry the End Cap role at the short sides of the floor plate. Source: AEC_Floor Plates_Tiles_Alternatives.pdf (V12, Collaborators #27).

TILE-A

Tile A — Corporate Office

2,700 SF / 250.84 m²

Single corporate tenant unit (2,700 SF).
Maximum climate-zone autonomy.

1 climate zone

TILE-B1

END CAP

Tile B-1 — Private Office End Cap

2,700 SF / 250.84 m²

Five private offices (300 + 450 + 300 + 450 + 300 SF) plus 900 SF corridor.

5 climate zones

TILE-C1

Tile C-1 — Professional (Med + Small)

2,700 SF / 250.84 m²

Professional Office Medium (2,000 SF) + Small (700 SF).

2 climate zones

TILE-C2

Tile C-2 — Professional (Med Pair)

2,700 SF / 250.84 m²

Two Medium Professional Office units (1,100 SF each) + 500 SF corridor.

2 climate zones

TILE-C3

Tile C-3 — Professional (Large+Small)

2,700 SF / 250.84 m²

Large Professional (2,400 SF) + Small Professional (300 SF). Lab/Medical anchor.

2 climate zones

TILE-C4

Tile C-4 — Professional Single Large

2,700 SF / 250.84 m²

Single 2,200 SF Professional unit + 500 SF corridor/support. Civic/Academic.

1 climate zone

TILE-E1

END CAP

Tile E-1 — Mixed End Cap (Left)

2,700 SF / 250.84 m²

2 × Private (300 SF) + Professional Small (800 SF) + 1,300 SF corridor.

3 climate zones

TILE-E2

END CAP

Tile E-2 — Mixed End Cap (Right)

2,700 SF / 250.84 m²

Mirror of E-1 for right building end. Handles corner column grid.

3 climate zones

The Tile Catalogue — Medium & Large Tiles

Three tile sizes form the catalogue: Small (2,700 SF), Medium (3,500 SF), Large (4,900 SF). Medium tiles bridge the gap between Small and Large; their End Cap variants (E-1, E-2) absorb the short-side geometry of the building. Source: AEC_Floor Plates_Methodology_V12.pdf p. 2.

TILE-D

MEDIUM

Tile D — Medium Corporate Office

3,500 SF / 325.16 m²

Single 3,500 SF corporate tenant. Medium-family corporate anchor.

1 climate zone

TILE-E

MEDIUM

Tile E — Medium Private Office

3,500 SF / 325.16 m²

Eight Private Offices (300–500 SF mix) totalling 2,900 SF + 600 SF corridor.

8 climate zones

TILE-FM

MEDIUM

Tile F (Medium) — Professional

3,500 SF / 325.16 m²

Four Professional Office units (3 Small 800 + 1 Medium 1,100). Medium-family professional anchor.

4 climate zones

TILE-F

LARGE

Tile F — Large Corporate Office

4,900 SF / 455.22 m²

Single 4,900 SF corporate tenant. Approaches 1/4 floor tile.

1 climate zone

TILE-G

LARGE

Tile G — Large Private Office Mix

4,900 SF / 455.22 m²

Ten Private Offices (300–500 SF mix) + 900 SF corridor. Maximum climate-zone count.

10 climate zones

TILE-H

LARGE

Tile H — Large Professional Office

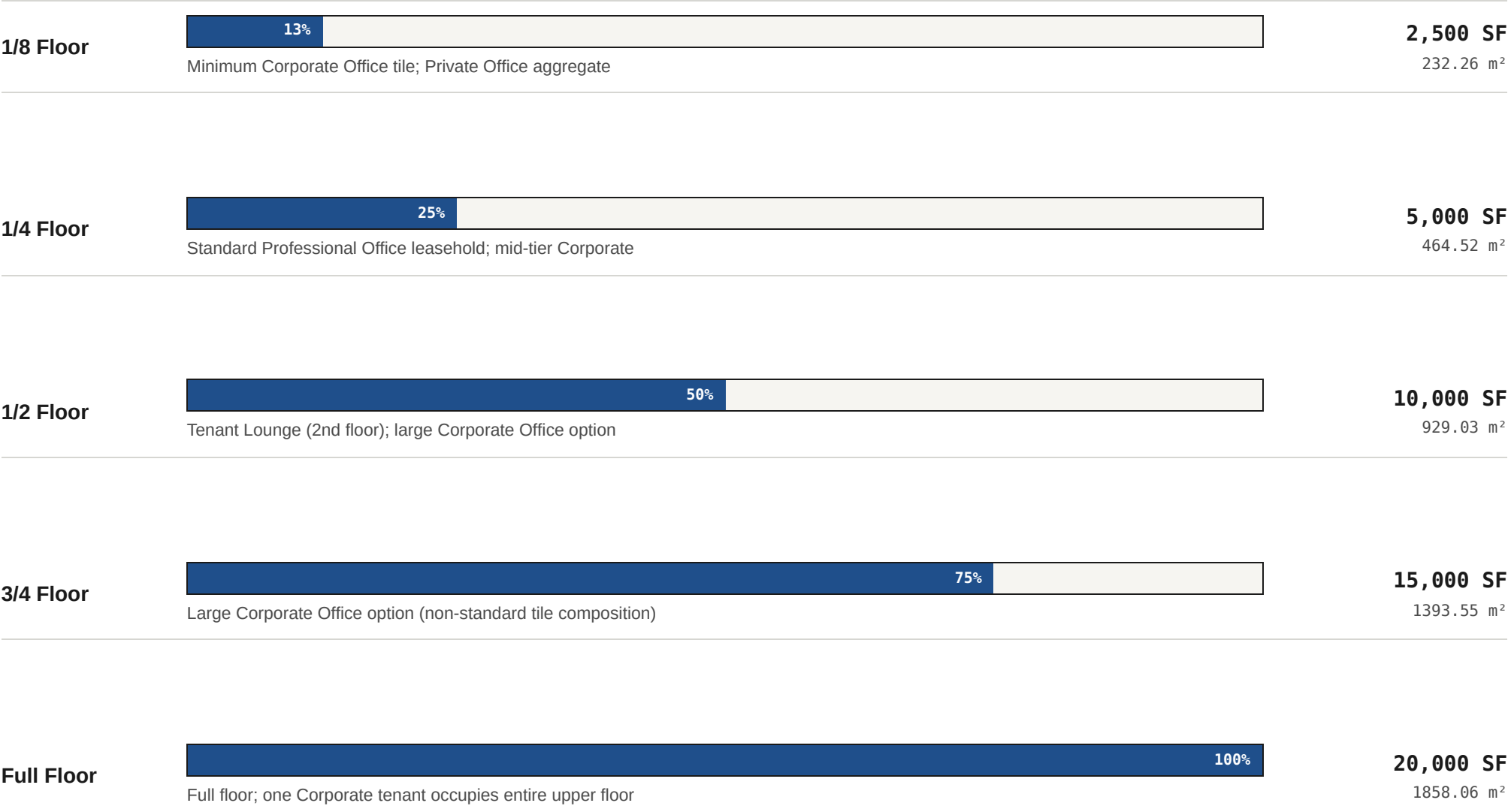
4,900 SF / 455.22 m²

Five Professional Office units (mixed Small/Medium/Large). Spans Medical, Business, Lab, Academic, Civic.

5 climate zones

The Size Hierarchy & Floor Plate Range

Five fractions of the reference 20,000 SF net leasable floor plate. The reference is a midpoint — Professional Centres run 19,000–23,000 SF and Suburban Office runs 17,000–21,000 SF, with 3–5 floors and 6–9 floors respectively. Source: PROJECTS_MCorp_Tear Sheet_Floor Plates_Tiles_Combinations.pdf p. 2.



A Corporate Office tenant may occupy any fraction up to the full floor. The 1/8 floor (2,500 SF) is the minimum Corporate Office tile *and* the aggregate tile for Private Office leaseholds. The 1/2 floor is dedicated on the second floor to the Tenant Lounge per the design memo.

One Floor — Filled Exactly

A worked composition — one Sample Small Tile arrangement from the methodology. The body of the floor is filled by Basic and Compound Tiles; the residual area around the Core and Elevator Lobby is filled by Special Tiles (SP-A, SP-B, SP-C). Together these sum to the full net leasable area with no remainder.

SAMPLE SMALL TILE – TILE A ×6 + SPECIAL A ×2 + SPECIAL B + CORE = 20,000 SF NET LEASABLE FOOTPRINT (V12 METHODOLOGY P. 4)



0 SF10,000 SF20,000 SF

- Tile A — 2,700 SF · 1 climate zone
- Tile A — 2,700 SF · 1 climate zone
- SP-A — 400 SF · 1 climate zone
- Core — 1,300 SF
- SP-B — 300 SF · 1 climate zone
- Tile A — 2,700 SF · 1 climate zone
- Tile A x2 — 5,400 SF · 2 climate zones
- SP-A — 400 SF · 1 climate zone
- Tile A — 2,700 SF · 1 climate zone
- Tile A — 1,400 SF · 1 climate zone

Other combinations are valid as long as they sum to net leasable with every SF represented by a Key Plan. *Special Tiles absorb the residual geometry around the building Core* — they are sized so their width matches the nearest Professional Office key-plan width, preserving demising-wall continuity. Source: V12 Methodology p. 4 (Sample Small Tile).

Design Principles

Encoded as token metadata at `bim.tile.design-principles`. Four principles govern every composition.

CLIMATE-ZONE AUTONOMY

One tile, one HVAC zone. A tenant who wants direct thermostat control must lease a whole Tile. Tenants combining tiles must combine climate control with the same negotiated authority. Smaller tiles = more zones = more autonomy per tenant; larger tiles = fewer zones = more contiguous tenant space.

ROLLING EFFICIENCY

The Tile composition must reach 100% floor-plate utilisation at any tenant mix — no free space without a corresponding Key Plan. As tenants come and go, the floor recomposes; the engineering rule is that the recomposition is always achievable from the published catalogue.

END CAPS & NATURAL LIGHT

Building short sides use End Cap tiles (B-1, E-1, E-2 at Small/Medium scale; A-1, B-2, C-3, C-4 at 6,000 SF for paired short sides). End caps must receive natural light on both perpendicular axes; if they do not, the leasehold is “not 100% efficient” per the methodology and a window-bay revision is triggered.

SPECIAL TILES SNAP TO KEY-PLAN WIDTH

Special Tiles SP-A, SP-B, SP-C fill the area around the Core and Elevator Lobby. Their width snaps to the nearest Professional Office Key Plan width — e.g. if a PO is 450 SF and SP-A is 400 SF, SP-A is increased by 50 SF to match. SP-C (in front of the Core) must avoid direct door-to-elevator alignment.

Leasing Economics & Rolling Efficiency

WHY SMALLER TILES MATTER

16 leasehold variations vs. 9

The methodology's quantitative test is leasehold variation count at a fixed plate area. A modular 325 SF grid yields nine leasehold sizes at 2,925 SF total. The Key Plans and Tiles system — using 325 SF, 465 SF, and 685 SF private-office widths in irregular combination — yields **sixteen leasehold sizes at 2,775 SF total**. More leasable configurations per square foot of plate. Source: Tiles Combinations p. 4.

Climate-zone autonomy is the second leasing lever. A small tenant who wants thermostat control must lease a full Tile. A Private Office in Tile B-1 shares climate with four other private offices; a Private Office in Tile G shares with nine; a Corporate Office in Tile A or F sets its own. **The leasing agreement prices climate autonomy explicitly.**

ROLLING EFFICIENCY

Engineering efficiency vs. economic efficiency

Engineering efficiency is net-leasable divided by gross-buildable — a static ratio. The Woodfine methodology replaces it with **economic efficiency**: the share of a floor that is actively leased and represented by a Key Plan matching a real tenant's furniture inventory.

Rolling efficiency is the dynamic test. As tenants come and go, the Tile recombination must always be valid; *the catalogue is sized so 100% utilisation is always recoverable* from a published combination. Tenants taking space across Tile boundaries lose climate-zone control — the leasing instrument prices this trade-off explicitly.